11. Which of these is not a monomer for a high molecular mass silicone polymer?

a) $Me_3SiCl$ b) $PhSiCl_3$ c) $MeSiCl_3$ d) $Me_2SiCl_2$

12. Which of the following is not $sp^2$ hybridised?

a) Graphite b) graphene c) Fullerene d) dry ice

13. The geometry at which carbon atom in diamond are bonded to each other is

a) Tetrahedral b) hexagonal c) Octahedral d) none of these

14. Which of the following statements is not correct?

a) Beryl is a cyclic silicate

b) $Mg_2SiO_4$ is an orthosilicate

c) $SiO_4^{4-}$ is the basic structural unit of silicates

d) Feldspar is not aluminosilicate

15. Match items in column - I with the items of column – II and assign the correct code.

| Column-I | | Column-II | |
|---|---|---|---|
| A | Borazole | 1 | $B(OH)_3$ |
| B | Boric acid | 2 | $B_3N_3H_6$ |
| C | Quartz | 3 | $Na_2[B_4O_5(OH)_4]\ 8H_2O$ |
| D | Borax | 4 | $SiO_2$ |

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 1 | 4 | 3 |
| (b) | 1 | 2 | 4 | 3 |
| (c) | 1 | 2 | 4 | 3 |
| (d) | None of these | | | |

16. Duralumin is an alloy of

a) Cu,Mn b) Cu,Al,Mg c) Al,Mn d) Al,Cu,Mn,Mg

17. The compound that is used in nuclear reactors as protective shields and control rods is

a) Metal borides b) metal oxides c) Metal carbonates d) metal carbide

18. The stability of +1 oxidation state increases in the sequence

a) Al < Ga < In < Tl b) Tl < In < Ga < Al

c) In < Tl < Ga < Al d) Ga< In < Al < Tl

## Answer the following questions:

1. Write a short note on anamolous properties of the first element of p-block.
2. Describe briefly allotropism in p- block elements with specific reference to carbon.
3. Give the uses of Borax.
4. What is catenation ? describe briefly the catenation property of carbon.
5. Write a note on Fisher tropsch synthesis.



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